AA–TV wiring v8.3: no wheels. 6x Deutsch DTP 4–pin (wet zone), servo leads + cables epoxy–potted into dryboxes, 14 AWG marine outside / 20 AWG on boards.

BUOY (surface, dry)

5V_PI
BT2
USB–C power bank 5 V / 3 A
GND_PI

J1
GPIO12/PWM0 — 5V — 2 — 5V_PI
GPIO13/PWM1 — 3V3 — 1
GPIO14/UART_TXD — ID_SD_I2C0/GPIO00 — 27
GPIO15/UART_RXD — ID_SC_I2C0/GPIO01 — 28
GPIO16/SPI1_CE2 — SDA_I2C1/GPIO02 — 3
GPIO17/SPI1_CE1 — SCL_I2C1/GPIO03 — 5
GPIO18/SPI1_CE0/PCM_CLK/PWM0
GPIO19/SPI1_MISO/PCM_FS — GPCLK0/GPIO04 — 7
GPIO20/SPI1_MOSI/PCM_DIN/PWM1 — GPCLK1/GPIO05 — 29
GPIO21/SPI1_SCLK/PCM_DOUT — GPCLK2/GPIO06 — 31
GPIO22/SDIO_CLK — CE1_SPI0/GPIO07 — 26
GPIO23/SDIO_CMD — CE0_SPI0/GPIO08 — 24
GPIO24/SDIO_DAT0 — MISO_SPI0/GPIO09 — 21
GPIO25/SDIO_DAT1 — MOSI_SPI0/GPIO10 — 19
GPIO26/SDIO_DAT2 — SCLK_SPI0/GPIO11 — 23
GPIO27/SDIO_DAT3
GND — 6
GND_PI
Raspberry Pi 4 (buoy)

Pi 4 talks to the Pico over USB serial (/dev/ttyACM0) and powers it from USB 5 V. Reset the Pico by power–cycling the Pi USB ports (uhubctl).

TETHER

W1
USB–A — micro–B
5V_PI — 1 — 5 — VBUS_PICO
GND_PI — 4 — 8 — GND
+ USB data
USB–A to micro–B, ~3 m

Vinyl air tube from the buoy; USB cable runs inside it. USB plugs are dry (buoy, Pico).

PEN1
A1: air–tube fitting (USB inside)

MAIN DRYBOX (upper, on hull)

BOARD A: Pico solder–board

R1 220 — GP12 — M4_PWM
R2 220 — GP13 — M5_PWM
R3 220 — GP14 — M6_PWM
R4 220 — GP15 — M7_PWM
R5 220 — GP16 — THR_PWM
R6 220 — GP19 — PUMP_PWM

VBUS_PICO — 39 — VSYS
40 — VBUS
+3V3 — 36 — 3V3
Raspberry Pi Pico
A1

RUN — 30
3V3_EN — 37
GPIO0 — 1
GPIO1 — 2
GPIO2 — 4
GPIO3 — 5
GPIO4 — 6
GPIO5 — 7
GPIO6 — 9
GPIO7 — 10
GPIO8 — 11
GPIO9 — 12
GPIO10 — 14
GPIO11 — 15
GPIO12 — 16
GPIO13 — 17
GPIO14 — 19
GPIO15 — 20
GPIO16 — 21 — GP16
GPIO17 — 22
GPIO18 — 24
GPIO19 — 25 — GP19
GPIO20 — 26
GPIO21 — 27
GPIO22 — 29
ADC_VREF — 35
GPIO26_ADC0 — 31 — LEAK_MAIN
GPIO27_ADC1 — 32 — LEAK_REAR
GPIO28_ADC2 — 34
AGND — 33 — GND
GND — 38
GND — 3

220 ohm in series on every PWM line that leaves the drybox.

TA2
1 — +3V3
2 — GND
3 — LEAK_MAIN
Main leak sensor

TA1
M4_PWM 1
M5_PWM 2
M6_PWM 3
M7_PWM 4
THR_PWM 5
H1_SPARE6 6
H1_SPARE7 7
PUMP_PWM 8
LEAK_REAR 9
+3V3 10
GND 11
GND 12
H1 out (14 AWG)

All Board A terminals: 5.08 mm pitch (0.2 in, fits the breadboard grid). TA1 pins 6–7: spare H1 conductors.

PEN2
A2: H1 cable, epoxy–potted

REAR DRYBOX (inside open rear endcap)

BOARD B: servo / signal board

TB1
M4_PWM 1
M5_PWM 2
M6_PWM 3
M7_PWM 4
THR_PWM 5
H1_SPARE6 6
H1_SPARE7 7
PUMP_PWM 8
LEAK_REAR 9
+3V3 10
GND 11
GND 12
H1 in (14 AWG)

TB7
1 SERVO_7V5
2 SERVO_7V5
3 GND
4 GND
7.5 V in (2x14 AWG per pole)

SERVO_7V5 — C1 1000uF 16V — GND

TB2
M4_PWM 1
SERVO_7V5 2
GND 3
M5_PWM 4
SERVO_7V5 5
GND 6
Rudder servo leads (potted)

TB3
M6_PWM 1
SERVO_7V5 2
GND 3
M7_PWM 4
SERVO_7V5 5
GND 6
Elevator servo leads (potted)

TB4
THR_PWM 1
GND 2
Thruster ESC signal (to D3)

TB5
PUMP_PWM 1
GND 2
Pump ESC signal (U3)

TB6
+3V3 1
GND 2
LEAK_REAR 3
Rear leak sensor

Board B:
– 7.5 V + GND rails: strip of 14 AWG soldered along strips
– Servo + / – jumpers: 14 AWG
– Signal jumpers: 20 AWG
– No 12 V on the breadboard
– TB1 pins 6–7: spare, unused

PEN3
R2: signal cables potted (H1, 4 servo leads, THR sig)

Off–board, in rear drybox

BB1
VBAT — 1 2 3 4 5 6
Bus bars (OONO 2x6 screw–clamp module, one row per pole, 30 A); in: D1 trunk, one 14 AWG per pole; out: buck U1, pump ESC U3, D3 thruster ESC feed (2 spare per row)
+12 V row (OONO 2x6, 30 A)

BB2
GND — 1 2 3 4 5 6
GND row (OONO 2x6, 30 A)

U1
VBAT 1 — VIN+ — VOUT+ — 3 SERVO_7V5
GND 2 — VIN– — VOUT– — 4 GND
BUCK
12 V –> 7.5 V buck (>=5 A)

U3
VBAT 1 — BAT+ — M+ — 6 PUMP_M+
GND 2 — BAT– — M– — 7 PUMP_M–
PUMP_PWM 4 — SIG
GND 5 — SIG_GND — ESC
Pump ESC (brushed, reversible)

Pump ESC sits dry, so any brushed ESC with reverse works. BEC wire cut. Motor leads –> R3 –> D2 pins 1–2 (extend with 14 AWG if short).

PEN4
R3: power cables potted (trunk, thruster feed, pump)

WET ZONE (flooded hull, open rear endcap, splash areas): Deutsch DTP 4–pin connectors live here

30 A ATC, waterproof holder (>=30 A)
F1
VBAT
BT1
3S LiPo (wet, no box)
SW1
GND
Kill switch, low side, >=30 A DC

D1A D1B
VBAT 1 1 VBAT
GND 2 2 GND
D1_LOOP 3 3 D1_LOOP
D1_LOOP 4 4 D1_LOOP
D1 battery DTP06–4S DTP04–4P
D1: 1 + / 2 GND (25 A contact each); 3–4 dummy loop.

Battery side: battery + –> F1 –> D1 pin 1; battery – –> SW1 –> D1 pin 2. D1 pins 3–4: dummy loop on each half (short 14 AWG piece, contact on each end) so every cavity is sealed. No splits. Charge through a DTP04–4P –> XT60 adapter.

D4A D4B
M4_PWM 1 1 M4_PWM
M5_PWM 2 2 M5_PWM
M6_PWM 3 3 M6_PWM
M7_PWM 4 4 M7_PWM
D4 H1 1–4 DTP06–4S DTP04–4P

D5A D5B
THR_PWM 1 1 THR_PWM
H1_SPARE6 2 2 H1_SPARE6
H1_SPARE7 3 3 H1_SPARE7
PUMP_PWM 4 4 PUMP_PWM
D5 H1 5–8 DTP06–4S DTP04–4P

D6A D6B
LEAK_REAR 1 1 LEAK_REAR
+3V3 2 2 +3V3
GND 3 3 GND
GND 4 4 GND
D6 H1 9–12 DTP06–4S DTP04–4P
H1 main <–> rear: 12 x 14 AWG over D4, D5, D6. Pins 8–7 spare.

M4_PWM 1
SERVO_7V5 2
GND 3
PWM
M4
Rudder servo
M5_PWM 1
SERVO_7V5 2
GND 3
PWM
M5
Rudder servo
M6_PWM 1
SERVO_7V5 2
GND 3
PWM
M6
Elevator servo
M7_PWM 1
SERVO_7V5 2
GND 3
PWM
M7
Elevator servo

Servos: 2x Hitec DB777WP (4.5 A stall @7.4 V), 2x Savox SW–1250MG (2.4 A). Servo leads: potted straight into the rear drybox at R2, no connector, BACKUP if a lead can't reach the box: one MGYOO 20 AWG 3–pin waterproof connector per servo, inline in the wet zone.

D3A D3B
VBAT 1 1 VBAT
GND 2 2 GND
THR_PWM 3 3 THR_PWM
GND 4 4 GND
D3 thruster DTP06–4S DTP04–4P
D3: 1 +12 V, 2 GND, 3 PWM (yellow), 4 signal GND (white). All 14 AWG.

U2
VBAT 1 — BAT+ — U — 5 — 1
GND 2 — BAT– — V — 6 — 2
THR_PWM 3 — SIG — W — 7 — 3
GND 4 — SIG_GND — BLDC ESC
TD1.2 ESC (waterproof, wet)
MT1 BLDC
TD1.2 thruster
Factory thruster cable stays on the ESC.

D2A D2B
PUMP_M+ 1 1 PUMP_M+
PUMP_M– 2 2 PUMP_M–
D2_LOOP 3 3 D2_LOOP
D2_LOOP 4 4 D2_LOOP
D2 pump DTP06–4S DTP04–4P
D2: 1 pump +, 2 pump –; 3–4 dummy loop.

PUMP_M+
M3
Ballast pump, 12 V brushed reversible (TBD, ~10 A)
PUMP_M–

Deutsch DTP: size 12 contacts, 12–14 AWG, 25 A each. Every cavity is filled: unused pins get a dummy wire loop. Source side (battery / drybox) gets the DTP06 plug with socket contacts.

Wire: 14 AWG marine for everything outside the boxes, every DTP contact and all power; 20 AWG hookup inside the boxes.
Only connectors: 6x Deutsch DTP 4–pin in the wet zone + 5.08 mm screw terminals / bus bars inside the dryboxes.
Two solder–boards (Board A main drybox, Board B rear drybox). Cables epoxy–potted through drybox walls.

PIE AA–TV
Sheet: /
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Title: AA–TV electrical v8.3: no wheels, 6x Deutsch DTP 4–pin, servos potted

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